Replication of "Too Young to Die: Deprivation Measures Combining Poverty and Premature Mortality"

Baland, Cassan and Decerf (2021), *American Economic Journal: Applied Economics*, 13(4), 226–257

Development Economics 2026
Madras School of Economics

Replication submission, 17 October 2026

[TEAM: names of the three members]

[TEAM: group leader, contact email and phone]

1 Introduction

[TEAM: Write this section yourselves. Points it must cover, with sources:]

- Puzzle: Botswana 1990–2000. Extreme poverty 33.6% → 29.5%, life expectancy 63.6 → 45.6 years (paper p. 226). Did total deprivation fall?
- Gap in existing measures: head-count ratio (HC) counts only the living. A poor person who dies leaves the denominator, so HC can fall because the poor die (the "mortality paradox", p. 231 fn 7).
- Contribution: dying before an age threshold $\hat{a}$ is a deprivation of its own (life-span deprivation). Person-years prematurely lost (PYPL) are added to person-years in alive deprivation (PYAD), then divided by one reference population. Two indexes: Inherited Deprivation (ID) and Generated Deprivation (GD).
- Theoretical mechanism to state explicitly (the rubric asks for it): premature death is exclusive of poverty (the dead cannot be poor), so the two deprivations add in time units. A single reference population removes the inconsistency of composite indexes (Table 1 below).
- Headline results to be replicated: ignoring mortality understates world deprivation by 27% (1990) to 36% (2015); GD and HC give opposite trend verdicts in 8.4% of five-year comparisons.
- What this report does: reproduces every table and figure except Figure 1 (a schematic), from the authors' public package, and explains each gap.
- Roadmap sentence.

2 Data

[TEAM: Write the data section. Facts to cover:]

- **Mortality and population:** Global Burden of Disease 2017 round (GBD Collaborative Network, 2018). All-cause deaths and rates per 100,000, both sexes, by age group, country and year, 1990–2017. Point estimates only; 5-year age bins except 0–1 and 1–4.
- **Alive deprivation:** PovCalNet (World Bank, 2019). Head-count ratio at $1.90 per day (2011 PPP). Survey-based, three-yearly, *linearly interpolated* to annual values (Chen & Ravallion, 2013).

- **Regions:** World Bank classification file.
- **Sample:** 113 low- and middle-income countries, 1990–2015, plus six regions and the World. A country enters only with an HC value in every year. The sample is 113 because the authors' name-based merge drops Eswatini ("Swaziland" in GBD) and West Bank and Gaza ("Palestine"), and Kiribati sits in their drop list; an ISO3 merge gives 116.
- **Vintage:** we use the exact files in the authors' openICPSR package (E119941V1), so no data revision enters. PovCalNet has since been retired (succeeded by PIP); later GBD rounds revise history.
- **France (Figure 5):** Human Mortality Database. **Not yet re-run from the shared repository** (files pending); see Section 5.

3 Methodology

[TEAM: Explain the two indexes in your own words. Equations and notation below are from the paper (pp. 232–234) and can be used as they stand.]

Notation: t is the year; $\hat{a}$ the life-span threshold; f, p the numbers alive non-poor and alive poor; d the number prematurely dead; n_a the number alive at age a; μ_a age-specific mortality; $\gamma > 0$ the weight of one PYPL relative to one PYAD.

$$ID_\gamma(x) = \frac{p(x)}{f(x)+p(x)+d(x)} + \gamma \frac{d(x)}{f(x)+p(x)+d(x)} \tag{1}$$

$$GD_\gamma(x,\mu) = \frac{p(x)}{f(x)+p(x)+d^{GD}(x,\mu)} + \gamma \frac{d^{GD}(x,\mu)}{f(x)+p(x)+d^{GD}(x,\mu)} \tag{2}$$

$$d^{GD}(x,\mu) = \sum_{a=0}^{\hat{a}-1} n_a(x)\,\mu_a\left(\hat{a}-(a+1)\right) \tag{3}$$

Things the methodology section must state:

- Baseline parameters: $\hat{a} = 50$, $\gamma = 1$, poverty line \$1.90. Other thresholds used: $\hat{a} = 5$ (Haiti, Figure 3), 40 (France, Figure 5), 80 (sensitivity, p. 251).
- **Timing convention.** Equation (3) books deaths at the end of the period (loss of $\hat{a}-(a+1)$ years). The authors' empirical code books them at the start (loss of $\hat{a}-a$ years), consistent with their footnote 3. Our empirical results use $\hat{a}-a$. The theory figures (Figures 2 and 4) use equation (3) as printed.

- Only GD is estimated for the world, because ID needs $\hat{a}$ years of past deaths by age. ID is computed only for Haiti (Figure 3) and France (Figure 5).
- How the code builds the panel: population at the start of the year $=$ deaths $\div$ (rate $/100{,}000$), truncated; "living population" is the end-of-year population (start minus deaths); 5-year bins are split into single ages assuming a constant death rate within a bin and geometrically declining cohort sizes; everyone in 95+ dies that year; regional and World aggregates sum deaths and population by age bin before the same split; HC is aggregated with population weights.
- Ranks and reversals: ranks ascending within a year (least deprived $= 1$); average $|\text{rank}_{GD} - \text{rank}_{HC}|$; a reversal is a country-year where GD_t/GD_{t-5} and HC_t/HC_{t-5} lie on opposite sides of 1.
- Software and workflow: Stata, written independently of the authors' do-files, then checked against the authors' own code run on the same raw data.

4 Replication

[TEAM: Introduce each result in one or two sentences of your own, say whether it matches, and point to Section 5 for gaps.]

4.1 Theory examples (no data)

Table 1. The composite index $P_{0.5}$ is inconsistent (paper p. 228)

	Society A	Society B	Society C
Head-count ratio (HC)	0.50	0.33	0.00
Share of young dead (LD)	0.00	0.25	0.50
Composite $P_{0.5}$	0.25	0.29	0.25

Source: our computation (`09_simulations.do`); matches the paper.

Table 2. HC against ID on distributions x, x', x'' (paper p. 231)

	x	x'	x''
HC	0	0	0.5
ID_1	0.5	0	0.5
ID_2	1.0	0	0.5

Source: our computation (`09_simulations.do`).

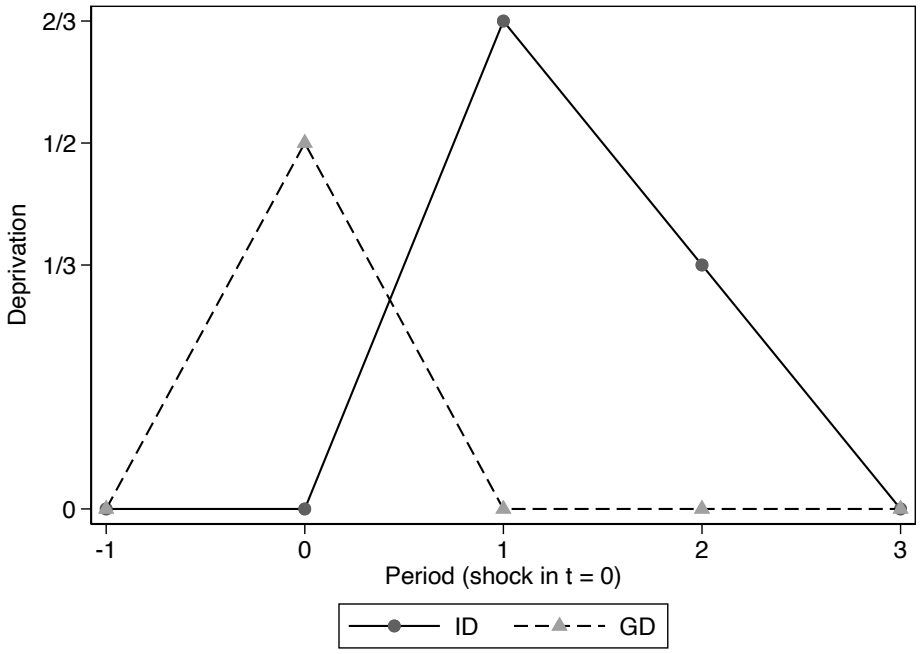


Figure 2. Response of ID and GD to a transitory shock. Source: our simulation.

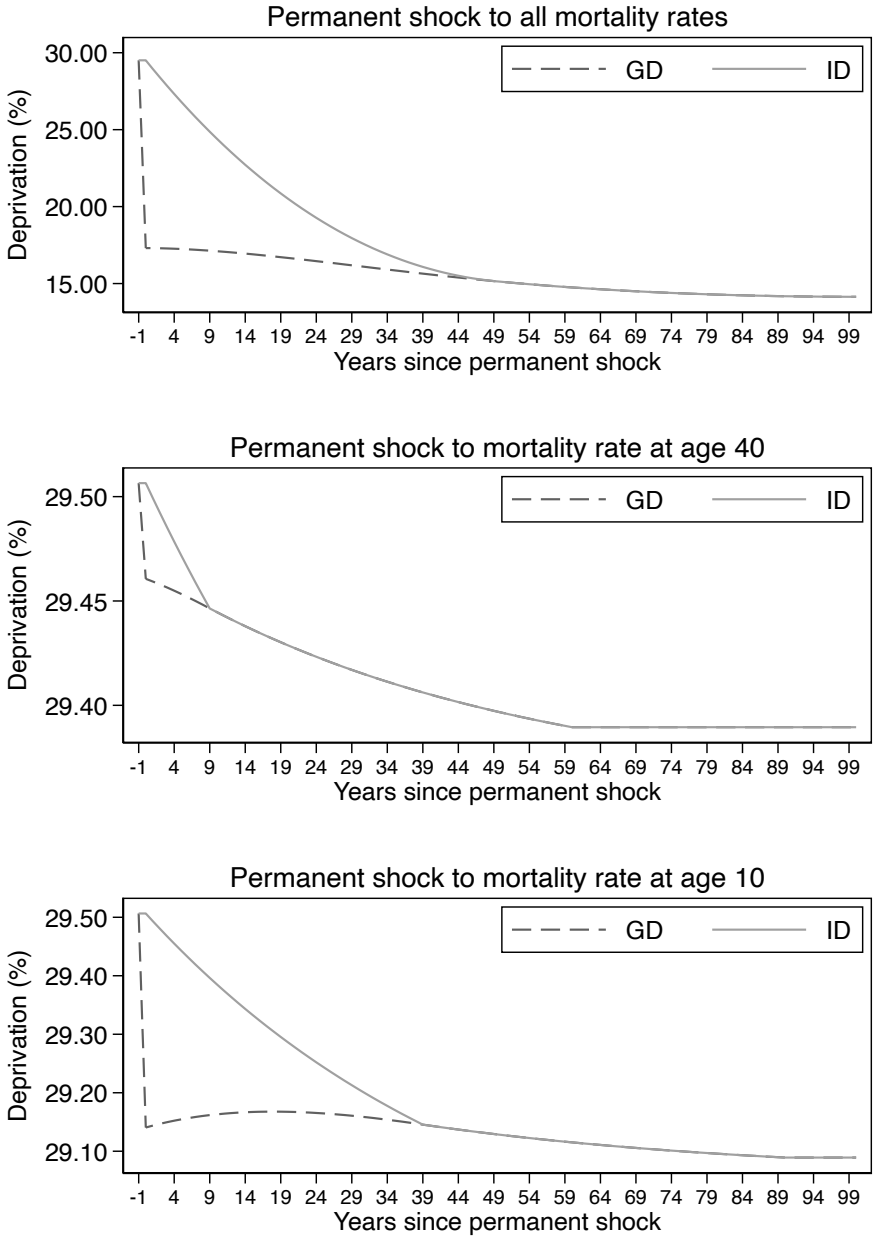


Figure 4. Response of ID and GD to permanent mortality shocks ($\hat{a} = 50$). Source: our simulation; identical to the authors' Excel sheet.

4.2 Worldwide deprivation

Table 3. World Generated Deprivation, $\hat{a} = 50$, $\gamma = 1$ (person-years in millions unless stated)

	1990		2015	
	Paper	Ours	Paper	Ours
Living population	4,200	4,197.9	6,010	6,007.0
Head-count ratio (%)	44.9	44.87	11.7	11.74
Alive deprivation (PYAD)	1,886	1,883.7	703	705.2
Life-span deprivation (PYPL)	701	701.4	402	402.4
Deprived population	2,587	2,585.1	1,105	1,107.6
Reference population	4,901	4,899.3	6,412	6,409.4
GD_1 (%)	52.8	52.76	17.2	17.28

Source: our computation (`10_tables.do`). The paper's cells are hand-computed from rounded rows (see Section 5).

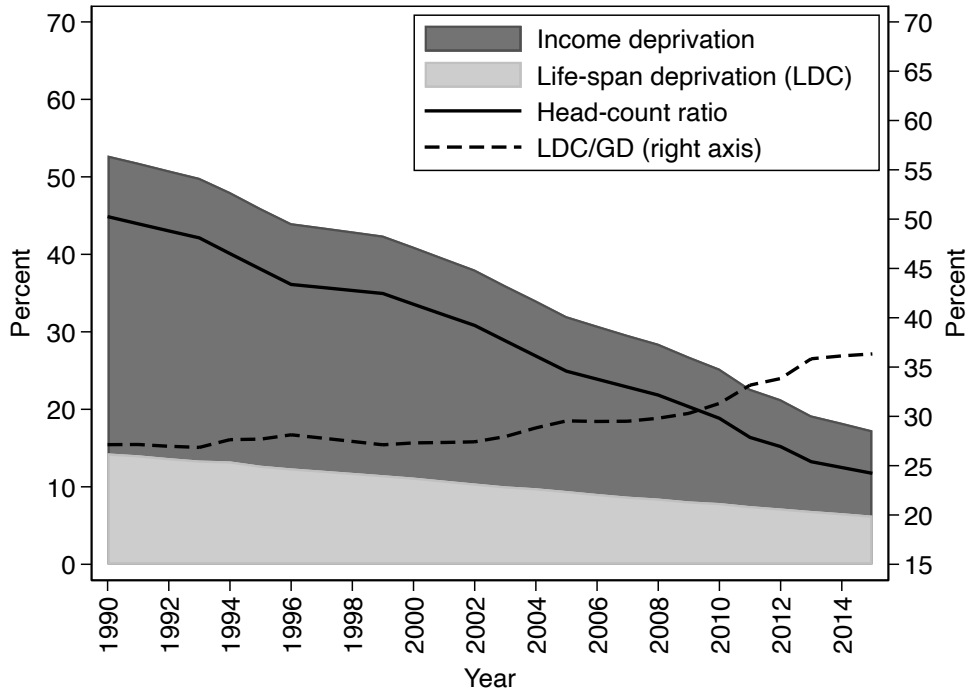


Figure 6. World deprivation: income and life-span components of GD, head-count ratio, and life-span share, 1990–2015. Source: our computation.

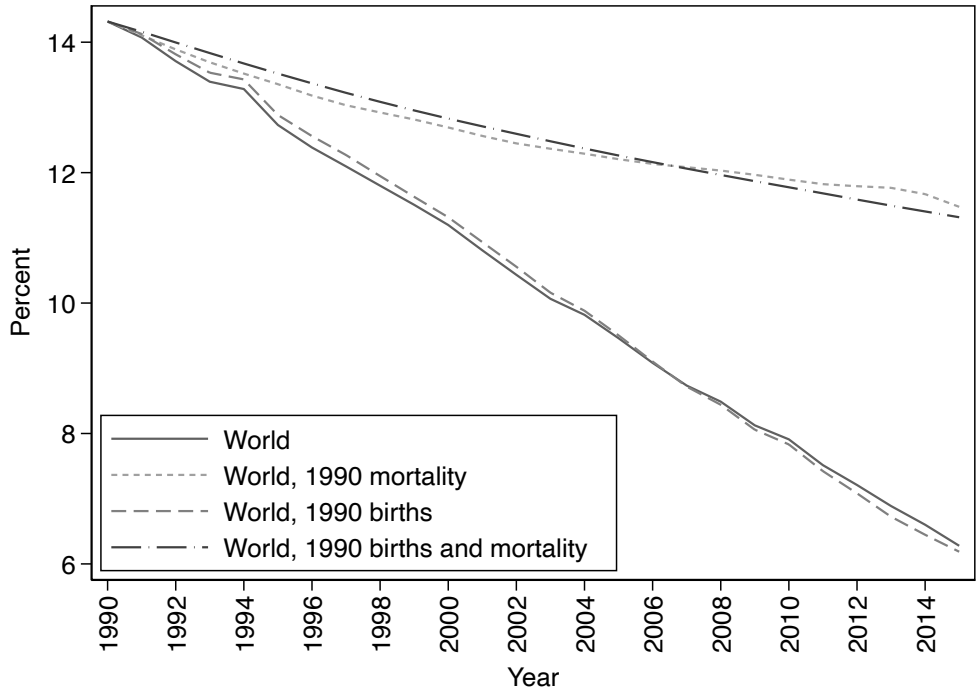


Figure 7. Life-span component of GD with 1990 births, 1990 mortality, or both held fixed. Source: our computation.

[TEAM: Say in your own words what Table 3 and Figures 6–7 show. Numbers to use: life-span share rose from 27.1% to 36.3%; at $\hat{a} = 80$ the 2015 share is 60.4%; under-5 deaths are 79% (1990) and 68% (2015) of PYPL; natality does not drive the fall (births +3.3%), mortality decline and pyramid composition do.]

4.3 Countries

Table 4. Gabon and Morocco, 1993 (person-years in thousands unless stated)

	Gabon		Morocco	
	Paper	Ours	Paper	Ours
Living population	1,043	1,043.8	26,449	26,449.0
Head-count ratio (%)	5.2	5.17	5.8	5.78
Alive deprivation (PYAD)	54	53.9	1,534	1,527.5
Life-span deprivation (PYPL)	174	173.9	3,100	3,100.3
Deprived population	228	227.8	4,634	4,627.8
Reference population	1,217	1,217.6	29,549	29,549.3
GD_1 (%)	**17.9**	**18.71**	15.7	15.66

Source: our computation (`10_tables.do`). Gabon's published GD_1 divides by 1,271 instead of 1,217.

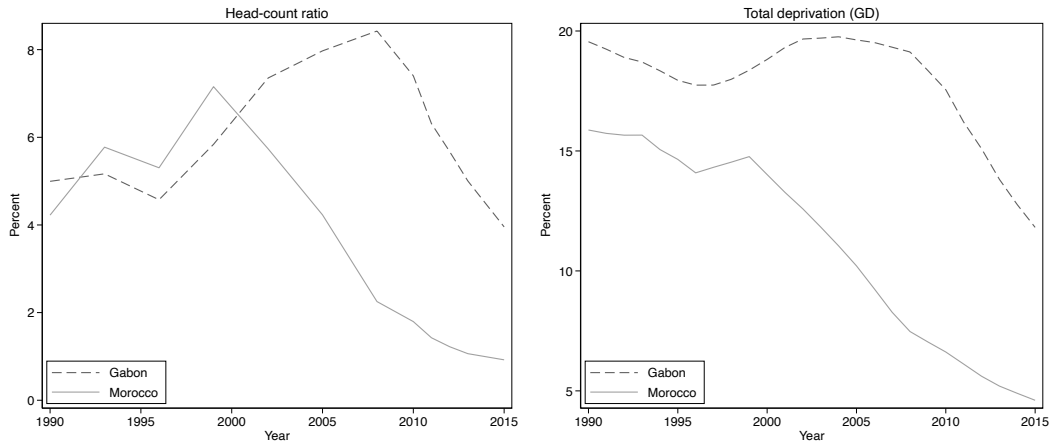


Figure 8. Gabon and Morocco: head-count ratio (left) and GD (right). Source: our computation.

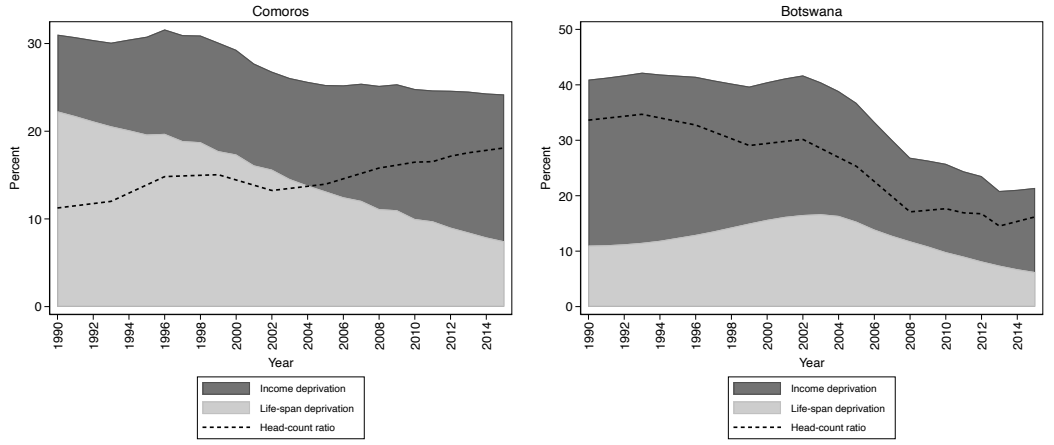


Figure 9. Comoros and Botswana: GD and HC, stacked. Source: our computation.

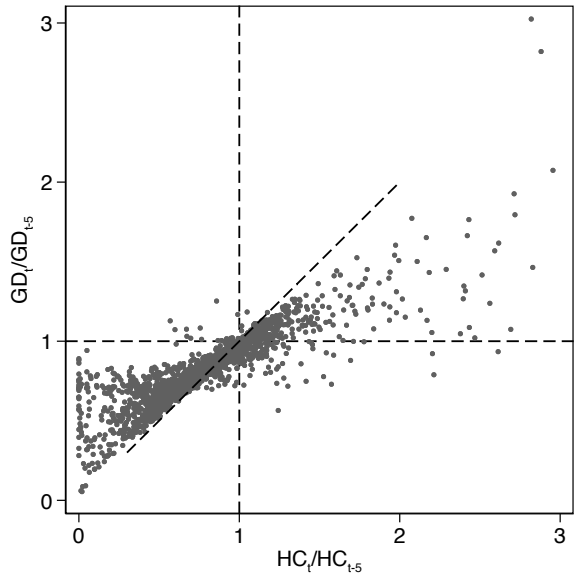


Figure 10. Five-year change in GD against five-year change in HC (HC ratio below 3). Source: our computation.

[TEAM: Say in your own words what Table 4 and Figures 8–10 show. Numbers to use: Gabon moves from rank 19 (HC) to 38 (GD), Morocco from 21 to 32 (113 countries, 1993); average rank change 4.1; Comoros HC +60.9% while GD -22.0%; Botswana HC -11.4% while GD $+0.5$% over 1990–2001; reversals 8.4% ($\gamma = 1$) and 26.9% ($\gamma \to \infty$).]

4.4 Haiti and France

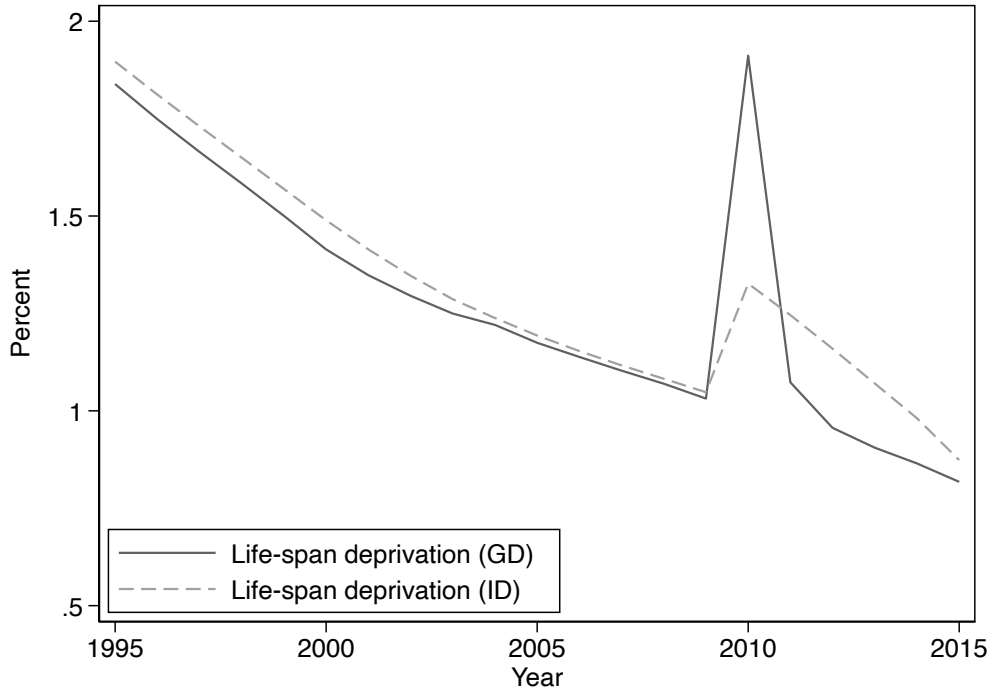


Figure 3. Haiti: life-span components of ID and GD, $\hat{a} = 5$, 1995–2015. Source: our computation.

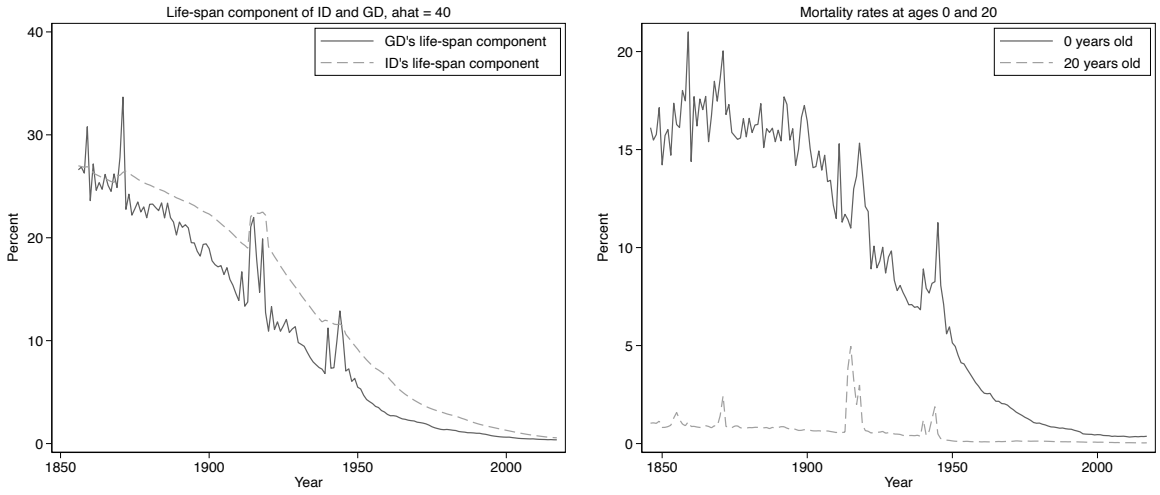


Figure 5. France: life-span components of ID and GD ($\hat{a} = 40$) and mortality at ages 0 and 20. Source: our computation from HMD period life tables (2026 vintage).

[TEAM: Figure 5 caveat: the HMD files were downloaded on 11 Oct 2026 by a teammate and are not in the shared repository. Do not describe Figure 5 as verified until the files are re-run (see audit note).]

5 Where our results differ from the published paper

[TEAM: Write this section in your own words; the rubric gives marks for explaining divergences. The table below lists every checked quantity that does not match, what the authors' own code gives, and the status. "Verified" means confirmed by running the authors' Stata code on their raw data.]

Table 5. Published numbers that do not match, and what explains them

Item	Paper	Ours	Authors' code	Status
Gabon GD_1, Table 4 (%)	17.9	18.71	18.71	Verified. Their do-file computes $228/1271$ (a typo for 1217); the correct value is 18.7.
Average rank change, 1990	4.4	4.92	4.92	Verified. The paper's 1990 and 2015 values are swapped.
Average rank change, 2015	4.9	4.39	4.34	Verified. Same swap. Our value differs from the authors' only through tie-breaking among countries with HC at or near zero.
Comoros HC change 1990–2015 (%)	+60	+60.9	same as ours	Within rounding.
Comoros GD change 1990–2015 (%)	-23	-22.0	same as ours	Unexplained. No start or end year in 1990–2015 gives -23.
Botswana HC change 1990–2001 (%)	-12	-11.4	same as ours	Sensitive to end year (-12.5 for 2000). Unexplained.
Botswana GD change 1990–2001 (%)	+0.7	+0.5	same as ours	No end year gives both published values. Unexplained.
Botswana HC in 2000 (%)	29.5	29.43	same as ours	Possibly the PovCalNet interpolation. Unexplained.
Newborn mortality 1990 and 2015 (%)	6.6, 3.4	6.76, 3.47	not computed	Definition not in the authors' code. Small gap.

Item	Paper	Ours	Authors' code	Status
Births, 1990–2015 (%)	+2.8	+3.3	not computed	Same.
Median age at death	64	58 (pooled), 65 (2015)	not computed	Possibly a comparison of different periods: the yearly median is 64–65 in 2012–2015. To be confirmed.

Further points the section should cover:

- **Rounding in the paper's Tables 3 and 4.** The authors' do-file builds those cells by hand from rounded rows (e.g. `di 26449*0.058` gives 1,534 for Morocco's PYAD, while the code's own value is 1,527.5). Applying the same rounding to our numbers reproduces every cell. Examples: the introduction's 705 million poor in 2015 is correct, and Table 3's 703 is $6{,}010 \times 0.117$.
- **Timing convention:** equation (3) ($\hat{a} - (a+1)$) against the code ($\hat{a} - a$). Resolved: use $\hat{a} - a$ for the empirics.
- **Sample:** 113 against 116 countries (Eswatini, West Bank and Gaza, Kiribati).
- **Ranks near the top of the table** are not robust: four countries have HC $= 0$ in 2015 and ten have HC $< 0.1\%$, so rank statistics depend on the tie rule. Azerbaijan's loss is 26 ranks in 2010 and 2014 under our rule; in the authors' output the maximum is 26 in 2008 and 25 in 2014.
- **Figure 5:** the package readme asks for cohort life tables, but the figure can only come from period tables (cohort tables stop at the 1933 cohort). The Alsace-Lorraine fall in ID appears in 1920 with 2026 HMD data, in 1919 in the paper. **[TEAM: State as unverified until the HMD files are re-run.]**
- **What matches exactly:** Table 3 (to rounding), life-span shares 27.1% and 36.3%, 60.4% at $\hat{a} = 80$, 113 countries, Gabon and Morocco ranks ($19 \to 38$, $21 \to 32$), reversals 8.42% and 26.88%, Botswana HC in 1990 (33.6%), Figure 4 against the authors' Excel.

6 Conclusion

[TEAM: Short conclusion: what replicated, what did not and why, and one sentence on the extension direction once the team has chosen it.]

Contribution of each member

[TEAM: Who did what (required in the final submission; keep a draft here).]

AI-use declaration

[TEAM: Declare, in the course's categories, what AI was used for. The AI-use log in the unit README lists what has been done so far.]

References

Baland, J.-M., Cassan, G., & Decerf, B. (2021). "Too young to die": Deprivation measures combining poverty and premature mortality. *American Economic Journal: Applied Economics, 13*(4), 226–257. https://doi.org/10.1257/app.20200122

Baland, J.-M., Cassan, G., & Decerf, B. (2021). *Replication data for: "Too young to die"* [Data set]. openICPSR. https://doi.org/10.3886/E119941V1

Chen, S., & Ravallion, M. (2013). More relatively-poor people in a less absolutely-poor world. *Review of Income and Wealth, 59*(1), 1–28. **[TEAM: verify against the source]**

GBD 2017 Collaborative Network. (2018). *Global Burden of Disease Study 2017 (GBD 2017) results*. Institute for Health Metrics and Evaluation. **[TEAM: add URL and access date]**

World Bank. (2019). *PovcalNet* [Database]. Development Research Group. **[TEAM: add URL and access date]**